

Decimal

To

Binary

Octal

HEX

Integer ↑
part

Fractional ↓
part

Int. ↑
part

Frac. ↓
part

Int. ↑
part

Frac ↓
part

Divide
by
2

Multiply
by
2

Divide
by
8

Multiply
by
8

Divide
by
16

Multiply
by
16

1. $D \xrightarrow{\div 2} 0$
 $\xleftarrow{\times 2}$

2. $D \xrightarrow{\div 8} 0$
 $\xleftarrow{\times 8}$

3. $D \xrightarrow{\div 16} H$
 $\xleftarrow{\times 16}$

For Integer Part

For Fraction, we multiply fractional part with respective radix, until it gets zero/repetative.

Decimal to Binary Conversion

$$r=2$$

>> To convert decimal to any other base 'r', divide integer part by r and multiply fractional part by r.

$$1) \quad (13)_{10} \rightarrow (1101)_2$$

$$13 = 8 + 4 + 1$$

1	1	0	1
b_3	b_2	b_1	b_0
↑	↑	↑	↑
2^3	2^2	2^1	2^0

$$2^0 = 1$$

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

$$\times 2^4 = 16 > 13$$

$$2) \quad (13)_{10} \rightarrow (1101)_2$$

Divisor	Dividend	
2	13	1 → rem
2	6	0
2	3	1
2	1	1
2	0	0
2	0	0

1101

msb



Ex.

$$\overset{I}{(25)} \cdot \overset{F}{(625)}_{10} \rightarrow (\quad)_2$$

i) $(67)_{10} \rightarrow (\quad)_2$

ii) $(129.75)_{10} \rightarrow (\quad)_2$

2	25	1	lsb ↑ msb
2	12	0	
2	6	0	
2	3	1	
2	1	1	
	0		

$$\underline{00}(11001.101)\underline{00}_2$$

$$\begin{aligned} 0.625 \times 2 &= 1.25 \\ 0.25 \times 2 &= 0.50 \\ 0.50 \times 2 &= 1.00 \\ 0.00 \times 2 &= 0.00 \end{aligned}$$

1. 1000011

2. 10000001.11

$$1) (112)_{10} \longrightarrow (160)_8$$

8	112	0	↑ lsb
8	14	6	
8	1	1	
	<u>0</u>		

msb

160

$$2) (25.625)_{10} \longrightarrow (31.50)_8$$

8	25	1	↑ lsb
8	3	3	
	0		

msb

31.50

***Decimal to Octal**

$$0.625 \times 8 = (5).000$$

$$0.000 \times 8 = (0).000$$

⋮



4.50. i) $(2048)_{10} \longrightarrow (\quad)_8$

$$0.000 \times 8 = \textcircled{0}.000$$

ii) $(82.250)_{10} \longrightarrow (\quad)_8$

⋮

1. 4000

2. 122.2

Decimal to Hexadecimal Conversion

$$\gamma = 16$$

>> For decimal to hexadecimal conversion, divide integer part by 16 and multiply fractional part by 16.

1) $(254)_{10} \longrightarrow (FE)_{16}$

$$16 \times 16 = 256$$

16	254	16 E
16	15	16 F
	0	

Labels:
 - $\swarrow$ lab (pointing to the first row)
 - $\searrow$ mab (pointing to the second row)

2) $(25.625)_{10} \longrightarrow (19.A)_{16}$

16	25	9
16	1	1
	0	

Label: $\uparrow$ (pointing to the second row)

$$0.625 \times 16 = \textcircled{A}.000$$

$$0.000 \times 16 = 0.000$$

0
...
9
A 10
B 11
C 12
D 13
E 14
F 15



$$3) (27.4)_{10} \rightarrow (123.1212)_4 \quad r=4$$

H.W.

$$(25.625)_{10} \rightarrow (\quad)_4$$

4	27	3	↑
4	6	2	
4	1	1	
	0		

msb
msb

$$0.4 \times 4 = \textcircled{1}.6$$

$$0.6 \times 4 = \textcircled{2}.4$$

$$0.4 \times 4 = \textcircled{1}.6$$

$$0.6 \times 4 = \textcircled{2}.4$$

$$\textcircled{1212}12 \dots$$

121.22